

Numerical methods for nonlocal conservation laws

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Preface

1 Introduction

Since many interesting conservation laws that arise in applications (e.g. traffic simulation) are nonlinear, it is impossible to obtain explicit solution formulas. Hence, we rely on designing numerical methods to approximate the solutions in the best possible way. When designing such methods it is natural to ask ourselves how reliable and efficient those methods could be. That is significant to study the stability and convergence of the numerical methods.

The main ingredient in designing numerical methods is first to study the properties of the analytical solutions. These properties are significant for validating the consistency and stability of the numerical methods. The well-posedness of the local conservation laws is stated in (Citation is needed) through viscous approximation. That means unique entropy solutions exist. Moreover the entropy solutions satisfy L^p stability estimates and are Total Variation Diminishing (TVD). Accordingly in (Citation is

needed) they designed conservative, consistent and monotone numerical scheme, such methods converge to the unique entropy solutions of the conservation laws as they preserve the stability estimates. The well-posedness of the nonlocal conservation laws (with non-local flux arising in traffic flow modeling) is shown in (Citation is needed), the proof is based on a priori L^∞ and (TVD) estimates of a numerical method. To obtain convergence of numerical methods for local conservative laws, it suffices to design conservative, consistent and monotone numerical schemes. However, for nonlocal conservation laws it's not a straight forward. In addition as nonlocal conservation laws are meant to be generalization of the local conservation laws, one wants to obtain the so-called asymptotically compatibility, that is as the parameters tend to zero one obtains the solution of the local conservation laws. It is therefore things get complicated when designing numerical methods for nonlocal conservation laws. They designed in (Citation is needed) a general class of numerical methods, that is a family of finite volume schemes. Under certain conditions such that the initial data can't have a negative jump, they proved the asymptotically compatibility and asymptotic preserving property. They provided in addition numerical experiments only for the Lax–Friedrichs type scheme. In the present work we provide further numerical experiments where we take a comparison of the Lax–Friedrichs type scheme and Godunov type scheme, and conclude that the Godunov type scheme is the efficient one.

Those scheme are however first order accurate. High-resolution schemes are needed in practice in order to increase the computational efficiency. A second order numerical scheme for the nonlocal conservation laws have been devolved in (Citation is needed).

Through out this thesis we focus on the specific one dimensional traffic flow model which is modeled by a nonlocal conservation law. In other words the goal is to design an efficient numerical method and ultimately to prove that the method convergences to the correct solution.

2 Preliminaries

2.1 Scalar local and nonlocal conservation laws

Scalar conservation laws are partial differential equations that can be written on the form

$$\begin{aligned}\partial_t u + \nabla \cdot f(u) &= 0 \\ u(x, 0) &= u_0(x)\end{aligned}\tag{1}$$

where $\partial_t = \frac{\partial}{\partial t}$ is temporal differentiation, $\nabla \cdot$ is divergence operator, $u = u(x, t)$ is the unknown function of x and t and f is a given flux function.

If we now let $f(u) = uv(u)$ where $v(u) = 1 - u$, then Eq. (1) can be used for modelling traffic flow. For a detailed setup of this model, we refer to [3]. This model is however based only on local information. when a car driver usually decides his speed, it depends on the traffic information within a road segment of length $\epsilon > 0$ ahead of the car's current location. For managing that case we look for a better way to model the velocity, so we need a nonlocal model which is on the following form

$$\begin{aligned}\partial_t u + \nabla \cdot (uV(u)) &= 0 \\ u(x, 0) &= u_0(x)\end{aligned}\tag{2}$$

where now V is a nonlocal operator, say,

$$\begin{aligned}V(u)(x, t) &= v(u * \omega_\epsilon) \\ &= 1 - \int_0^\epsilon \omega_\epsilon(y) u(x + y, t) dy \\ &= 1 - \int_x^{x+\epsilon} \omega_\epsilon(y - x) u(y, t) dy \quad \text{Change of variable}\end{aligned}\tag{3}$$

for some nonlocal kernel $\omega_\epsilon : [0, \epsilon] \rightarrow \mathbb{R}$ satisfying $\omega_\epsilon \geq 0$ and $\int_0^\epsilon \omega_\epsilon(y) dy = 1$. Formally speaking, $V(u) \rightarrow v(u)$ as $\epsilon \rightarrow 0$, but whether the corresponding solutions u_ϵ of (2) converge to a solution of (1), is the key question that has been studied in the literature (Citation is needed here).

In conclusion, let us denote (1) and (2) the local and nonlocal model, respectively. We are then aiming to approximate solutions of the local model by solutions of the nonlocal model, where the solution $u(x, t)$ represents the density of cars which is the number of cars per square meter.

2.2 Solutions to the Riemann problem

For numerical experiments we need entropy solutions for different initial data. Hence, we briefly mention the notion of weak solution and entropy solution.

Definition 2.1 (Weak solution) *A function $u \in L^\infty(\mathbb{R} \times \mathbb{R}_+)$ is a weak solution of (1) with initial data $u_0 \in L^\infty(\mathbb{R})$ if the following identity holds for all test functions $\phi \in C^1(\mathbb{R} \times \mathbb{R}_+)$*

$$\int_{\mathbb{R}_+} \int_{\mathbb{R}} u \phi_t + f(u) \phi_x \, dx \, dt + \int_{\mathbb{R}} u_0 \phi(x, 0) \, dx = 0. \quad (4)$$

Since weak solutions are not unique, some additional conditions have to be imposed in order to pick a weak solution that describes the physical flow correctly. One such condition is entropy condition. Detailed properties of the entropy solution can be found in [4]. We now look at the explicit solutions for the Riemann problem. The Riemann problem is the initial value problem

$$\begin{aligned} \partial_t u + \nabla \cdot f(u) &= 0 \\ u(x, 0) &= \begin{cases} u_l & \text{if } x < 0 \\ u_r & \text{if } x > 0 \end{cases} \end{aligned} \quad (5)$$

Solution of this equation is constructed as

$$u(x, t) = \begin{cases} u_l & \text{for } x \leq h'(u_l)t \\ (h')^{-1}\left(\frac{x}{t}\right) & \text{for } h'(u_l)t \leq x \leq h'(u_r)t \\ u_r & \text{for } x \geq h'(u_r)t \end{cases} \quad (6)$$

where $h(u)$ is defined by

$$h(u) = \begin{cases} \sup\{g(u) \mid g \leq f \text{ and } g \text{ is convex on } [u_l, u_r]\} & \text{if } u_l < u_r, \\ \inf\{g(u) \mid g \geq f \text{ and } g \text{ is concave on } [u_r, u_l]\} & \text{if } u_r < u_l. \end{cases} \quad (7)$$

3 Numerical Methods

3.1 Finite Volume Scheme for a General flux f

Designing a finite difference method typically consists of four steps:

1. Discretizing the domain,
2. Satisfying the equation at discrete points,
3. Replacing derivatives by finite differences,
4. Solving the discretized problem.

However, step 3 requires that solutions are sufficiently smooth, but solutions of conservation laws may not be differentiable or even continuous. In addition, the unknown solution u of a conservation law may represent a density (e.g., the average number of cars per m^2), in this case step 2 is not flexible. For these reasons, we must adapt the steps above to design a suitable numerical method.

3.2 Discretizing the Domain and Control Volumes

For simplicity, we consider a uniform discretization of both the one-dimensional spatial domain $[x_L, x_R]$ and the temporal domain $[0, T]$. We split the spatial domain $\Omega = [x_L, x_R]$ into K smaller subdomains Ω^j , such that

$$\begin{aligned} \Omega &= \bigcup_{j=0}^{K-1} \Omega^j \\ &= \bigcup_{j=0}^{K-1} [x_L + j\Delta x, x_L + (j+1)\Delta x], \end{aligned} \quad (8)$$

where $\Delta x = \frac{x_R - x_L}{K}$.

We denote the midpoint of each Ω^j by

$$x_j = x_L + \left(j + \frac{1}{2}\right)\Delta x, \quad j = 0, \dots, K-1,$$

and use each midpoint value to define the computational cells or control volumes:

$$\mathcal{C}_j = [x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}),$$

where the left interface $x_{j-\frac{1}{2}}$ and the right interface $x_{j+\frac{1}{2}}$ of $\mathcal{C}_j$ are given by

$$x_{j-\frac{1}{2}} = x_j - \frac{\Delta x}{2}, \quad x_{j+\frac{1}{2}} = x_j + \frac{\Delta x}{2}.$$

We have now split Ω into K smaller non-overlapping subdomains called control volumes $\mathcal{C}_j$. These control volumes are essential for the finite volume method, as pointwise evaluation does not make sense.

Setting $\Delta t = \frac{T}{N}$, we also discretize the temporal domain $[0, T]$ as

$$0 = t^0 < t^1 < \dots < t^N = T, \quad \text{where } t^n = n\Delta t, \quad n = 0, \dots, N.$$

3.3 Cell Averages and a Finite Volume Scheme

As solutions of conservations laws may be discontinuous, pointwise evaluation does not make sense. Instead, at each time level t^n , we take cell averages:

$$u(x_j, t^n) = u_j^n \approx \frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u(x, t^n) dx \quad (9)$$

Assuming now the cell averages are known at some time level t^n , we find the cell averages at next time level t^{n+1} as follows:

Taking an integral of (1) over the domain $[x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}] \times [t^n, t^{n+1}]$

$$\int_{t^n}^{t^{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_t dx dt + \int_{t^n}^{t^{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} f(u)_x dx dt = 0$$

And applying the fundamental theorem of calculus yields (**Clarification is needed on why this theorem is applicable**)

$$\begin{aligned} & \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u(x, t^{n+1}) dx - \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u(x, t^n) dx \\ &= - \int_{t^n}^{t^{n+1}} f(u(x_{j+\frac{1}{2}}, t)) dt + \int_{t^n}^{t^{n+1}} f(u(x_{j-\frac{1}{2}}, t)) dt \end{aligned} \quad (10)$$

Defining the numerical fluxes

$$\bar{F}_{j+\frac{1}{2}} = \frac{1}{\Delta t} \int_{t^n}^{t^{n+1}} f(u(x_{j+\frac{1}{2}}, t)) dt \quad (11)$$

and dividing both sides of (10) by Δx , we obtain

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (\bar{F}_{j+\frac{1}{2}} - \bar{F}_{j-\frac{1}{2}}) \quad (12)$$

Note that the relation in (12) is not explicit, since $\bar{F}$ requires a priori knowledge of the exact solution. By introducing an appropriate approximation of the numerical flux $\bar{F}$ by F , we obtain the finite volume scheme for a general flux f :

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}) \quad (13)$$

3.4 Lax–Friedrichs type Scheme and Godunov type scheme

Let the flux be defined as $f = uV$, where V is the nonlocal operator given in (3). There are several ways to specify the nonlocal numerical fluxes. In this work, we focus on Lax–Friedrichs type scheme and Godunov type scheme; see the approaches in [1] and [2].

If we define the numerical flux in (13) as

$$F_{j+\frac{1}{2}} = \frac{1}{2}(u_j^n V_j^n + u_{j+1}^n V_{j+1}^n) + \frac{\alpha}{2}(u_j^n - u_{j+1}^n) \quad (14)$$

where $\alpha > 0$ is the numerical viscosity, we establish a Lax–Friedrichs type scheme designed for the nonlocal model (2).

Mark that V_j^n is the approximation of of the nonlocal density $V(u)$ given by

$$V_j^n = 1 - \Delta x \sum_{k=0}^{m-1} w_k u_{j+k}^n, \quad (15)$$

Where $\{w_k\}_{k=0}^{m-1}$ is a set of numerical quadrature weights that are used to approximate the nonlocal kernel ω_ϵ and $m = \lceil \frac{\epsilon}{\Delta x} \rceil$ is the number of cells involved in V .

The numerical quadrature weights w_k involved in (15) can be given by left endpoint

$$w_k = \omega_\epsilon(k\Delta x)\Delta x \text{ for } k = 0, \dots, m-1,$$

normalized left endpoint

$$w_k = \frac{\omega_\epsilon(k\Delta x)\Delta x}{\sum_{k=0}^{m-1} \omega_\epsilon(k\Delta x)\Delta x} \text{ for } k = 0, \dots, m-1$$

and exact quadrature

$$w_k = \int_{k\Delta x}^{\min\{(k+1)\Delta x, \epsilon\}} \omega_\epsilon(y) dy \text{ for } k = 0, \dots, m-1$$

Moreover, if we let the numerical flux as

$$F_{j+\frac{1}{2}} = u_j^n V_{j+1}^n \quad (16)$$

We obtain a Godunov type scheme. One may ask whether the choice of numerical quadrature will result in different outcomes when implementing the schemes developed above. That is, whether the schemes converges for some numerical quadrature weights but fails to converge for others.

This question is addressed in [1] through numerical experiments and mathematical explanations. We will return to this question in the next section, but for now we consider two main aspects related to the finite volume scheme: the boundary conditions and stability.

3.5 The Boundary Conditions

The algorithm for solving (2) using for instance the The Lax–Friedrichs type scheme becomes:

STEP 1. Initialize u_j^0 by setting:

$$u_j^0 = \frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_0(x) dx, \quad \forall j \in \{0, \dots, K-1\}.$$

STEP 2. for $n = 0, \dots, N-1$, compute:

$$\begin{aligned} u_j^{n+1} &= u_j^n - \frac{\Delta t}{\Delta x} (F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}) \\ &= \frac{1}{2}(u_{j+1}^n V_{j+1}^n - u_{j-1}^n V_{j-1}^n) - \frac{\alpha}{2}(u_{j+1}^n + u_{j-1}^n) + \alpha u_j^n. \end{aligned}$$

The second step is also supposed to hold for all $j \in \{0, \dots, K-1\}$. However, for $j = 0$ we obtain:

$$u_0^{n+1} = \frac{1}{2}(u_1^n V_1^n - u_{-1}^n V_{-1}^n) - \frac{\alpha}{2}(u_1^n + u_{-1}^n) + \alpha u_0^n,$$

and for $j = K - 1$:

$$u_{K-1}^{n+1} = \frac{1}{2}(u_K^n V_K^n - u_{K-2}^n V_{K-2}^n) - \frac{\alpha}{2}(u_K^n + u_{K-2}^n) + \alpha u_{K-1}^n.$$

We observe that for $j = 0$ and $j = K - 1$, the scheme involves undefined terms such as u_{-1} , V_{-1} , u_K , and V_K . Therefore, without additional information the scheme is valid only for internal nodes $j \in \{1, \dots, K - 2\}$.

The above discussion indicate that specifying boundary conditions are needed in order to find the values u_0^{n+1} and u_{K-1}^{n+1} .

There are several options for specifying the boundary conditions on the spatial domain $[x_L, x_R]$, but the choice depends on whether the local model (1) is posed as an initial value problem (IVP) or a boundary value problem (BVP).

- **Initial Value Problem (IVP):** Artificial boundary conditions are required to handle the truncated domain $[x_L, x_R]$. Truncation is necessarily for initial value problem on the entire real line $\mathbb{R}$ as computational domain needs to be bounded.
- **Boundary Value Problem (BVP):** Common choices include Dirichlet, Neumann, and Periodic boundary conditions.

In this work, we focus on **artificial boundary conditions**. For a detailed discussion of the remaining types, see [Citation needed here](#). In this case, we set the values u_0^{n+1} and u_{K-1}^{n+1} to

$$u_0^{n+1} = u_1^{n+1} \quad \text{and} \quad u_{K-1}^{n+1} = u_{K-2}^{n+1}$$

That means the complete algorithm for solving (2) using the Lax–Friedrichs type scheme becomes:

STEP 1. Initialize u_j^0 by setting:

$$u_j^0 = \frac{1}{\Delta x} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u_0(x) dx, \quad \forall j \in \{0, \dots, K - 1\}.$$

STEP 2. For $n = 0, \dots, N - 1$ and $j = 0, \dots, K - 1$, compute:

$$\begin{aligned} u_j^{n+1} &= u_j^n - \frac{\Delta t}{\Delta x} (F_{j+\frac{1}{2}} - F_{j-\frac{1}{2}}) \\ &= \frac{1}{2}(u_{j+1}^n V_{j+1}^n - u_{j-1}^n V_{j-1}^n) - \frac{\alpha}{2}(u_{j+1}^n + u_{j-1}^n) + \alpha u_j^n. \end{aligned}$$

set

$$u_0^{n+1} = u_0^n \quad \text{and} \quad u_{K-1}^{n+1} = u_{K-2}^n$$

3.6 The Stability

We must make sure that the approximation solution obtained from whether the Lax–Friedrichs type scheme or Godunov type scheme shares similar properties as the entropy solution. Since the schemes may devolve undesired oscillations, we must impose a condition on the parameters Δx , Δt and α . So, we choose the parameters such that

$$\Delta t \leq \frac{\Delta x}{\alpha + 2} \quad \text{for Lax–Friedrichs} \tag{17}$$

and

$$\Delta t \leq \Delta x \quad \text{for Godunov} \tag{18}$$

in order to obtain stability. In order to understand the stability conditions deeply, see [Citation is needed here](#).

4 Numerical Experiments

In the previous section we have devolved the Lax–Friedrichs type scheme and the Godunov type Scheme for the nonlocal model. We described in particular the complete algorithm on how to implement the Lax–Friedrichs type scheme. In this section we consider if the numerical solutions converge to the correct solution numerically.

4.1 Comparison of Left endpoint and Normalized Left endpoint

We start by comparing the quadrature weights obtained from the left endpoint and the normalized left endpoint. For the Riemann problem (5), we choose $u_L = 0.6$ and $u_R = 0.1$ such that the problem becomes to solve

$$\begin{aligned} u_t + f(u)_x &= 0 \\ u(x, 0) &= \begin{cases} 0.6 & \text{if } x < 0 \\ 0.1 & \text{if } x > 0 \end{cases} \end{aligned} \quad (19)$$

Since $f(u) = u(1 - u)$ is a convex function, h in (7) equals f itself. Thus, the entropy solution is given by:

$$u(x, t) = \begin{cases} 0.6 & \text{for } x \leq -0.2t, \\ \frac{t - x}{2t} & \text{for } -0.2t \leq x \leq 0.8t, \\ 0.1 & \text{for } x \geq 0.8t. \end{cases} \quad (20)$$

We plot u at final time $T = 1$ over the domain $[x_L, x_R] = [-2, 2]$ together with both the local and nonlocal numerical approximations. The linear decreasing kernel $\omega_\epsilon(y) = \frac{2(\epsilon - y)}{\epsilon^2}$ is used and the numerical viscosity constant α is set to 2.

Numerical experiment 1a:

We begin by implementing the Lax–Friedrichs type scheme for $\epsilon = 0.1$ and $\Delta x = 0.02$

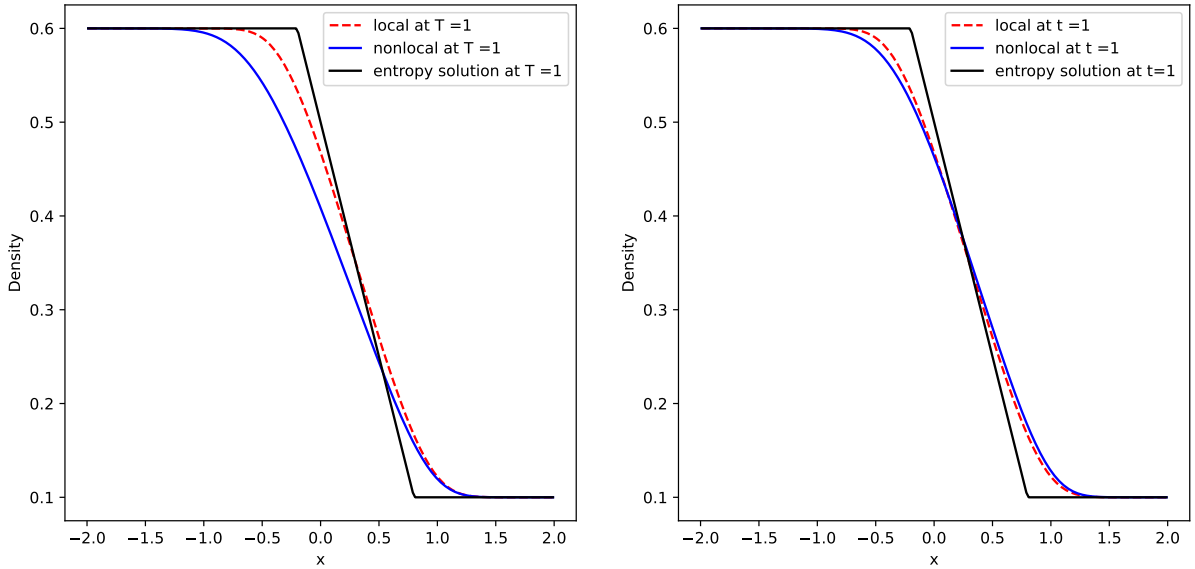


Figure 1: Numerical solutions and entropy solution at $T = 1$ corresponding to the left endpoint numerical quadrature weights (left) and the normalized left endpoint numerical quadrature weights (right).

As shown in Figure 1, the choice of numerical quadrature weights does not lead to a significant difference between the numerical solutions and the entropy solution. Nevertheless, a low level of accuracy remains at the shocks, where the numerical solutions deviate noticeably from the entropy solution. We aim to achieve convergence of the numerical solutions to the entropy solution as Δx and ϵ tend to zero simultaneously.

Numerical experiment 1b:

Next we examine the implementation for $\epsilon = 0.01$ and $\Delta x = 0.01$

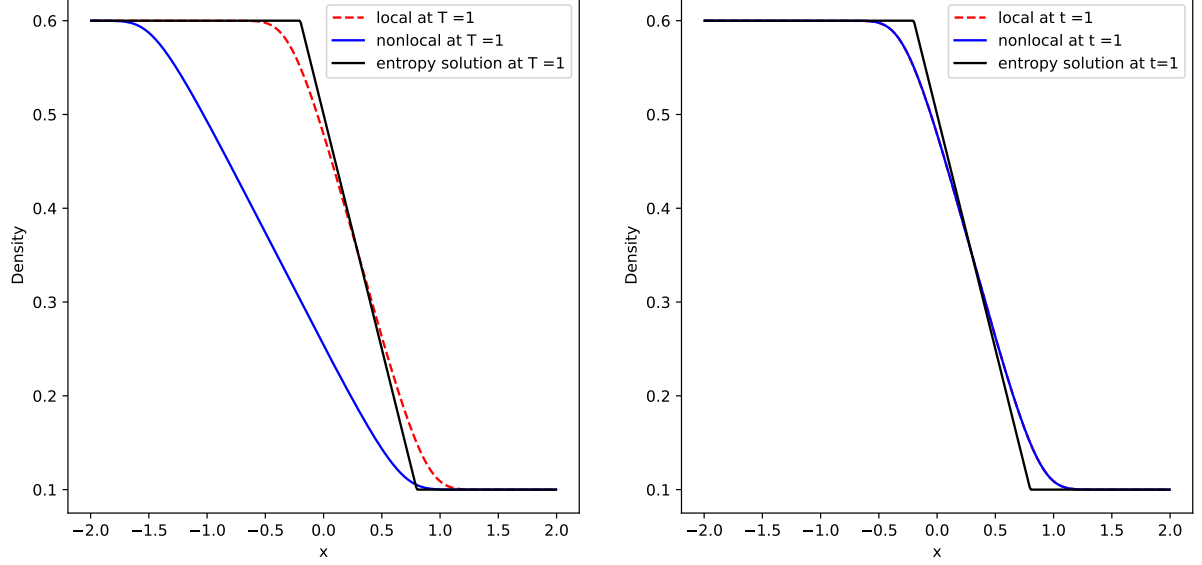


Figure 2: Numerical solutions and entropy solution at $T = 1$ corresponding to the left endpoint numerical quadrature weights (left) and the normalized left endpoint numerical quadrature weights (right).

As observed in Figure 2, contrary to our expectations, the left endpoint numerical quadrature weights cause the numerical solution of the nonlocal model to deviate further from both the numerical solution of the local model and the entropy solution. In contrast, the normalized left endpoint quadrature weights lead to improved accuracy and better alignment between the numerical solutions and the entropy solution.

Numerical experiment 1c:

Finally taking $\epsilon = 0.001$ and $\Delta x = 0.004$, the numerical solution of the nonlocal model obtained using the left endpoint numerical quadrature weights completely fails. By contrast, significantly better accuracy is observed when using the normalized left endpoint numerical quadrature weights.

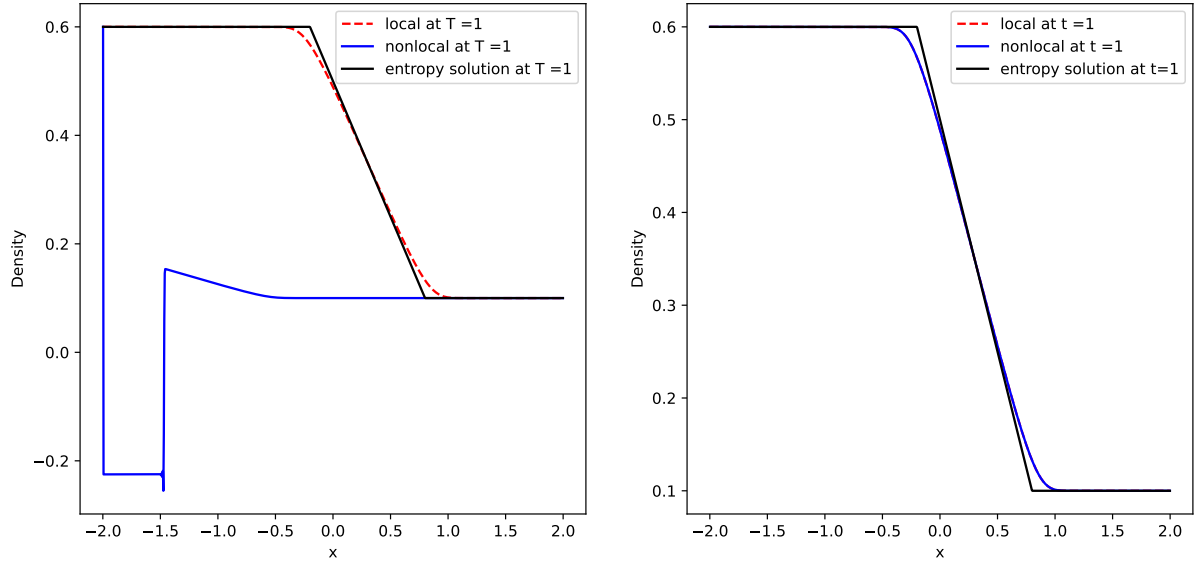


Figure 3: Numerical solutions and entropy solution at $T = 1$ corresponding to the left endpoint numerical quadrature weights (left) and the normalized left endpoint numerical quadrature weights (right).

Summary.

The numerical experiments above demonstrate that, as Δx and ϵ tend to zero simultaneously, while the Lax–Friedrichs type scheme based on the left endpoint quadrature weights becomes increasingly inaccurate and eventually fails to produce a meaningful numerical solution, the normalized left endpoint quadrature weights consistently maintain stability and improved alignment with the entropy solution. These results

indicate that the choice of numerical quadrature weights plays a crucial role in ensuring consistency between the Lax–Friedrichs scheme and the local model. The same applies for the Godunov type scheme as well, it turns out that for the sake of consistency it's required that the numerical quadrature weights must satisfy the normalized condition $\sum_{k=0}^{m-1} w_k = 1$.

4.2 Numerical Convergence Analysis

Based on the numerical experiments and arguments we provided above, both the Lax–Friedrichs scheme and Godunov scheme are consistent and stable when using the normalized left endpoint quadrature weights. Let us continue the numerical experiment to determine the convergence and order of convergence, and see if one scheme is more accurate than the other.

Denote:

- U^0 : the reference solution of the local model.
- $U^{0,\Delta x}$: the numerical solution of the local model.
- U^ϵ : the reference solution of the nonlocal model.
- $U^{\epsilon,\Delta x}$: the numerical solution of the nonlocal model.

We are interested in presenting numerical experiments on whether:

- for fixed ϵ , $U^{\epsilon,\Delta x} \rightarrow U^\epsilon$ as $\Delta x \rightarrow 0$.
- $U^{\epsilon,\Delta x} \rightarrow U^0$ as both $\Delta x \rightarrow 0$ and $\epsilon \rightarrow 0$ simultaneously.

Considering the computed reference and numerical solutions as piecewise constant functions on each cell of size Δx , we measure the size of the global error $e = U^{ref} - U^{num}$ using the L^1 norm:

$$\|e\|_{L^1} = \Delta x \sum_{j=0}^{K-1} |e^j|.$$

where U^{ref} and U^{num} corresponding to the the reference and numerical solution respectively. We desire to get $\|e\|_{L^1} \rightarrow 0$ as the parameters tend to zero.

4.3 Numerical Convergence to the Nonlocal Model

We begin investigating if the devolved numerical schemes converge numerically to the solution of the nonlocal model. For fixed ϵ , through numerical experiments we aim to deduce $\|U^\epsilon - U^{\epsilon,\Delta x}\|_{L^1} = \|e\|_{L^1} \rightarrow 0$ as $\Delta x \rightarrow 0$.

Since the analytic solution is not always available, it is practical to compute the reference solution numerically on a very fine grid. Focusing on the initial data given in (19), we compute the reference solution U^ϵ for $\Delta x = \frac{x_R - x_L}{4096} = \frac{4}{4096}$ and $\epsilon = 0.1$. Holding $\epsilon = 0.1$ fixed and letting $\Delta x \rightarrow 0$, although $\Delta x = \frac{1}{2^{4+m}}$ for $m = 0, 1, 2, 3, 4, 5, 6$, we compute the L^1 errors and plot against the number of cells to compare the efficiency of the Godunov scheme and Lax–Friedrichs scheme. We also preset the observed order of convergence in Table 1

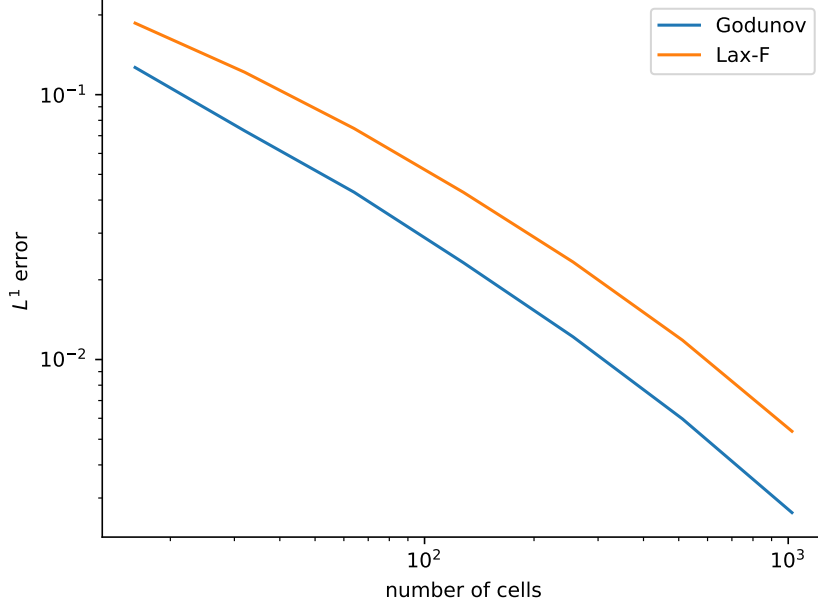


Figure 4: L^1 error vs. number of cells

Figure 4 indicates that the L^1 -error for both Godunov and Lax–Friedrichs scheme tends to zero as the number of cells increases. In other words, for fixed ϵ both the schemes converge to the reference (weak) solution U^ϵ of the nonlocal model as $\Delta x \rightarrow 0$. We observe as well that the Godunov scheme converges faster than Lax–Friedrichs scheme. The reason lies on how the schemes are devolved, The Godunov scheme doesn't involve the numerical viscosity constant α at all, while the Lax–Friedrichs scheme does, which results in less efficiency. A natural question is: how fast do the schemes converge? To answer this, we study the order of convergence r for both schemes.

We assume that the L^1 error on a mesh of size Δx_i can be written as

$$E_i = C(\Delta x_i)^r,$$

for some constant $C > 0$. If we have the error at two consecutive refinement levels (E_{i-1}, E_i) , then

$$\frac{E_{i-1}}{E_i} = \left(\frac{\Delta x_{i-1}}{\Delta x_i} \right)^r.$$

Taking logarithms, we obtain the standard expression for the order of convergence r :

$$r = \frac{\log(E_{i-1}) - \log(E_i)}{\log(\Delta x_{i-1}) - \log(\Delta x_i)}.$$

The order of convergence tells us how fast does a scheme converge to the correct solution, the higher order is the faster convergence.

Cells	L^1 -errors	OOC
16	1.8626×10^{-1}	–
32	1.2159×10^{-1}	0.6154
64	7.4547×10^{-2}	0.7058
128	4.2751×10^{-2}	0.8022
256	2.3350×10^{-2}	0.8725
512	1.1829×10^{-2}	0.9811
1024	5.3589×10^{-3}	1.1423

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	1.2667×10^{-1}	–
32	7.3044×10^{-2}	0.7942
64	4.2832×10^{-2}	0.7701
128	2.3189×10^{-2}	0.8853
256	1.2184×10^{-2}	0.9285
512	5.9654×10^{-3}	1.0303
1024	2.6423×10^{-3}	1.1748

(b) Godunov scheme

Table 1: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the Riemann initial data (19).

Table 1 presents the quantitative results for the L^1 -errors and the observed order of convergence r for the Riemann initial data (19), where $u_L > u_R$. The corresponding entropy solution is a rarefaction wave, see (20). We observe that the L^1 -errors produced by the Godunov scheme are consistently smaller than those obtained using the Lax–Friedrichs scheme. For both schemes, the observed order of convergence is at most 1, indicating relatively slow convergence.

Let us instead choose the Riemann initial data $u_L = 0.1$ and $u_R = 0.6$ such that $u_L < u_R$, the entropy solution is given by

$$u(x, t) = \begin{cases} 0.1 & \text{if } x < 0.3t \\ 0.6 & \text{if } x > 0.3t \end{cases}$$

which contains a single shock, and we obtain the corresponding quantitative results:

Cells	L^1 -errors	OOC
16	2.0628×10^{-1}	–
32	1.0767×10^{-1}	0.9380
64	7.1225×10^{-2}	0.5962
128	3.8519×10^{-2}	0.8868
256	2.0491×10^{-2}	0.9106
512	1.0478×10^{-2}	0.9676
1024	4.8977×10^{-3}	1.0972

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	1.0844×10^{-1}	–
32	4.4663×10^{-2}	1.2798
64	3.3717×10^{-2}	0.4056
128	1.6571×10^{-2}	1.0248
256	8.2085×10^{-3}	1.0134
512	3.9258×10^{-3}	1.0641
1024	1.7225×10^{-3}	1.1885

(b) Godunov scheme

Table 2: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the Riemann initial data $u_L = 0.1 < 0.6 = u_R$

The observation in Table 2 indicates that both the numerical schemes solve effectively solutions with shocks than rarefactions, **why?**, in the sense that the L^1 -errors and the order of convergence (in particular for the Godunov scheme) are improved. If we repeat the above experiments, but with the smooth initial data:

$$u(x, 0) = 0.5 + 0.3 \cos \frac{\pi x}{2} \quad (21)$$

we obtain

Cells	L^1 -errors	OOC
16	3.3092×10^{-1}	–
32	1.8065×10^{-1}	0.8733
64	9.9861×10^{-2}	0.8552
128	5.2189×10^{-2}	0.9362
256	2.6458×10^{-2}	0.9801
512	1.2647×10^{-2}	1.0649
1024	5.5133×10^{-3}	1.1978

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	1.8322×10^{-1}	–
32	8.6063×10^{-2}	1.0901
64	4.5197×10^{-2}	0.9291
128	2.2487×10^{-2}	1.0071
256	1.1061×10^{-2}	1.0237
512	5.1421×10^{-3}	1.1050
1024	2.2161×10^{-3}	1.2143

(b) Godunov scheme

Table 3: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the smooth initial data (21).

We observe in Table 3 that the L^1 -errors are significantly improved compared to the L^1 -errors for the Riemann initial data. But we still observe that both the schemes are first order accurate. This behavior is expected, since both the Lax–Friedrichs scheme and Godunov scheme are based on piecewise constant approximations and are therefore first order accurate.

Summary.

The quantitative results above indicate that for fixed ϵ , $U^{\epsilon, \Delta x} \rightarrow U^\epsilon$ as $\Delta x \rightarrow 0$. In addition, the Godunov scheme is more efficient than the Lax–Friedrichs scheme. Now it is time to consider the quantitative results as both the parameters ϵ and Δx tend to zero.

4.4 Numerical Convergence to the Local Model

Rarefaction solution

Cells	L^1 -errors	OOC
16	2.5205×10^{-1}	–
32	1.8141×10^{-1}	0.4744
64	1.2020×10^{-1}	0.5938
128	7.7851×10^{-2}	0.6267
256	4.9129×10^{-2}	0.6641
512	3.0257×10^{-2}	0.6993
1024	1.8184×10^{-2}	0.7346

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	2.0744×10^{-1}	–
32	1.4154×10^{-1}	0.5515
64	9.0816×10^{-2}	0.6402
128	5.7350×10^{-2}	0.6632
256	3.5334×10^{-2}	0.6987
512	2.1299×10^{-2}	0.7303
1024	1.2565×10^{-2}	0.7614

(b) Godunov scheme

Table 4: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the local model with the Riemann initial data (19).

Discontinuous solution

Cells	L^1 -errors	OOC
16	1.2876×10^{-1}	–
32	8.5352×10^{-2}	0.5932
64	4.9126×10^{-2}	0.7969
128	2.6836×10^{-2}	0.8723
256	1.3336×10^{-2}	1.0089
512	6.7539×10^{-3}	0.9815
1024	3.3509×10^{-3}	1.0112

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	1.2876×10^{-1}	–
32	8.5331×10^{-2}	0.5935
64	4.9126×10^{-2}	0.7966
128	2.5173×10^{-2}	0.9646
256	1.2981×10^{-2}	0.9555
512	5.3551×10^{-3}	1.2774
1024	2.3655×10^{-3}	1.1787

(b) Godunov scheme

Table 5: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with the Riemann initial data $u_L = 0.1 < 0.6 = u_R$

Smooth solution

Cells	L^1 -errors	OOC
16	4.4054×10^{-1}	–
32	2.7531×10^{-1}	0.6782
64	1.6123×10^{-1}	0.7719
128	9.0594×10^{-2}	0.8317
256	4.9131×10^{-2}	0.8828
512	2.5444×10^{-2}	0.9493
1024	1.2219×10^{-2}	1.0581

(a) Lax–Friedrichs scheme

Cells	L^1 -errors	OOC
16	3.1430×10^{-1}	–
32	1.8983×10^{-1}	0.7275
64	1.0959×10^{-1}	0.7926
128	6.1083×10^{-2}	0.8433
256	3.3015×10^{-2}	0.8876
512	1.7170×10^{-2}	0.9432
1024	8.4186×10^{-3}	1.0283

(b) Godunov scheme

Table 6: L^1 -errors and observed order of convergence obtained using the the Lax–Friedrichs scheme and the Godunov scheme in solving the nonlocal model with smooth initial data (21).

5 A Second order Numerical Scheme

5.1 Semidiscrete Formulation

The integral form of the conservation law we stated in section 3 is suitable for first order accurate schemes. In order to drive higher order schemes, we need a more flexible formulation which is called a semidiscrete formulation. This formulation allows us to separately increase the order of spatial and temporal accuracy, see [Citation needed here](#).

Taking now each cell average over C_j to be given by

$$u_j(t) = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t) dx \quad (22)$$

and integrating (1) over the domain C_j

$$\int_{x_{j-1/2}}^{x_{j+1/2}} u_t + f(u)_x dx = 0$$

gives us

$$\frac{d}{dt} u_j(t) + \frac{1}{\Delta x} \left(f(u(x_{j+1/2}^-, t)) - f(u(x_{j-1/2}^+, t)) \right)$$

where

$$u(x_{j+1/2}^-, t) = \lim_{x \rightarrow x_{j+1/2}^-} u(x, t)$$

and

$$u(x_{j+1/2}^+, t) = \lim_{x \rightarrow x_{j+1/2}^+} u(x, t)$$

denotes the left and right-limit of u at $x_{j+1/2}$, respectively.

Let F be an approximation of the fluxes such that,

$$F_{j+1/2}^\pm \approx f(u(x_{j+1/2}^\pm, t)),$$

then we obtain the semidiscrete formulation as

$$\begin{cases} \frac{d}{dt} u_j(t) + \frac{1}{\Delta x} (F_{j+\frac{1}{2}}^- - F_{j-\frac{1}{2}}^+) = 0 \\ u_j(0) = \int_{x_{j-1/2}}^{x_{j+1/2}} u_0 dx \end{cases} \quad (23)$$

Which is a system of ODEs that must be integrated in time by a suitable time integration routine.

5.2 Second Order Semidiscrete Scheme

Having the semidiscrete formulation (23), we can now separately increase the order of spatial and temporal accuracy. We observe in the formulation above that we need the edge values $u_j^\pm = u(x_{j+1/2}^\pm, t)$. If we let the edge values to be piecewise constants in space and apply a forward Euler time integration, we obtain the standard first-order finite volume scheme. In order to obtain a second-order finite volume scheme, we need to construct a linear piecewise function in each cell, so the edge values are evaluated on the reconstructed functions.

Given a cell average u_j^n of C_j at time t^n , we can construct a linear function

$$\bar{u}_j(x) = u_j^n + \sigma_j \frac{x - x_j}{\Delta x}, \quad x \in C_j$$

where the slope σ_j is selected, such that $\bar{u}_j(x)$ is conservative and satisfy the TVD property.

We choose the slope by the minmod limiter

$$\sigma_j^n = \minmod \left(u_j^n - u_{j-1}^n, \frac{1}{2}(u_{j+1}^n - u_{j-1}^n), u_{j+1}^n - u_j^n \right).$$

Here

$$\minmod(a_1, \dots, a_m) = \begin{cases} \text{sgn}(a_1) \min_{1 \leq \ell \leq m} |a_\ell|, & \text{if } \text{sgn}(a_1) = \dots = \text{sgn}(a_m), \\ 0, & \text{otherwise.} \end{cases}$$

We can now use $\bar{u}_j$ to approximate $u(x_{j+1/2}^-, t)$ and $u(x_{j+1/2}^+, t)$, such that the left and right-limit values are given by

$$\begin{aligned} u_{j+\frac{1}{2}}^- &:= \lim_{x \rightarrow x_{j+1/2}^-} \bar{u}_j(x) \\ &= u_j^n + \frac{1}{2} \sigma_j^n \end{aligned}$$

and

$$\begin{aligned} u_{j+\frac{1}{2}}^+ &:= \lim_{x \rightarrow x_{j+\frac{1}{2}}^+} \bar{u}_j(x) \\ &= u_{j+1}^n - \frac{1}{2} \sigma_{j+1}^n \end{aligned}$$

In section 3 we approximated the convolution term

$$\int_0^\epsilon \omega_\epsilon(y) u(x+y, t) dy$$

using piecewise constants, which led us to the first-order finite volume scheme. In order to obtain the second-order finite volume scheme, we approximate the convolution term using the piecewise linear function $\bar{u}(x, t) = \bar{u}_j(x)$ for $x \in C_j$ and trapezoidal rule as following:

Following the approach in [Citation needed here](#), we assume that the function ω_ϵ is a convolution kernel with compact support in $[0, \epsilon]$.

let

$$M := \lceil \frac{\epsilon}{\Delta x} \rceil,$$

and

$$I_{j+1/2} := \int_0^\epsilon \omega_\epsilon(y) u(x_{j+1/2} + y, t) dy.$$

We apply first $\bar{u}$ such that $u(x_{j+1/2} + y, t) \approx \bar{u}(x_{j+1/2} + y, t)$ to get

$$\begin{aligned} I_{j+1/2} &\approx \int_0^\epsilon \omega_\epsilon(y) \bar{u}(x_{j+1/2} + y, t) dy \\ &= \sum_{k=0}^{M-1} \int_{k\Delta x}^{(k+1)\Delta x} \omega_\epsilon(y) \bar{u}(x_{j+1/2} + y, t) dy. \end{aligned}$$

We can now apply the trapezoidal rule to get

$$\begin{aligned} I_{j+1/2} &\approx \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(\omega_\epsilon(k\Delta x) \bar{u}(x_{j+1/2} + k\Delta x, t) + \omega_\epsilon((k+1)\Delta x) \bar{u}(x_{j+1/2} + (k+1)\Delta x, t) \right) \\ &= \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(\omega_\epsilon(k\Delta x) \bar{u}(x_{j+k+1/2}, t) + \omega_\epsilon((k+1)\Delta x) \bar{u}(x_{j+k+1+1/2}, t) \right) \\ &= \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(w_k \bar{u}(x_{j+k+1/2}, t) + w_{k+1} \bar{u}(x_{j+k+3/2}, t) \right) \\ &= \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(w_k u_{j+k+1/2}^+ + w_{k+1} u_{j+k+3/2}^- \right) \end{aligned}$$

where w_k is the left end point numerical quadrature denoted by

$$w_k = \omega_\epsilon(k\Delta x), \quad k = 0, \dots, M$$

We have seen in section 3 that the normalization of the numerical quadrature plays essential role in consistency between the local and nonlocal model. Hence normalization is needed here as well.

Define

$$Q_{\Delta x} := \frac{\Delta x}{2} \sum_{k=0}^{M-1} (w_k + w_{k+1}), \quad \tilde{w}_k := \frac{w_k}{Q_{\Delta x}}, \quad k = 0, \dots, M$$

So the normalization condition is satisfied, in the sense

$$\frac{\Delta x}{2} \sum_{k=0}^{M-1} (\tilde{w}_k + \tilde{w}_{k+1}) = 1.$$

If we now replace w_k by $\tilde{w}_k$ in

$$R_{j+1/2} = \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(w_k u_{j+k+1/2}^+ + w_{k+1} u_{j+k+3/2}^- \right),$$

we get the approximation of the convolution term

$$R_{j+1/2} = \frac{\Delta x}{2} \sum_{k=0}^{M-1} \left(\tilde{w}_k u_{j+k+1/2}^+ + \tilde{w}_{k+1} u_{j+k+3/2}^- \right).$$

Letting the numerical flux $F_{j+1/2}^-$ in (23) be defined by the Godunov-type flux

$$F_{j+1/2}^- := u_{j+1/2}^- (1 - R_{j+1/2}),$$

and denoting

$$L_\epsilon(u)_j = \frac{1}{\Delta x} (F_{j+1/2}^- - F_{j-1/2}^+),$$

we obtain the semidiscrete nonlocal scheme:

$$\begin{cases} \frac{d}{dt} u_j(t) + L_\epsilon(u)_j = 0 \\ u_j(0) = \int_{x_{j-1/2}}^{x_{j+1/2}} u_0(x) dx \end{cases} \quad (24)$$

Proposition 5.1 *Let $\Delta x > 0$. Then, as $\epsilon \rightarrow 0$, the nonlocal semidiscrete scheme (24) converges to the local semidiscrete scheme*

$$\begin{cases} \frac{d}{dt} u_j(t) + L(u)_j = 0, \\ u_j(0) = \int_{x_{j-1/2}}^{x_{j+1/2}} u_0(x) dx, \end{cases} \quad (25)$$

where

$$L(u)_j := \frac{1}{\Delta x} (G_{j+1/2}^- - G_{j-1/2}^+), \quad G_{j+1/2}^- := u_{j+1/2}^- (1 - u_{j+1/2}^+)$$

5.3 Comparison of First Order and Second Order Godunov Scheme

5.4 Numerical Convergence to the Nonlocal Model

Rarefaction solution

Cells	L^1 -errors	OOC
16	1.2667×10^{-1}	—
32	7.3044×10^{-2}	0.7942
64	4.2832×10^{-2}	0.7701
128	2.3189×10^{-2}	0.8853
256	1.2184×10^{-2}	0.9285
512	5.9654×10^{-3}	1.0303
1024	2.6423×10^{-3}	1.1748

(a) First order Godunov scheme

Cells	L^1 -errors	OOC
16	5.0110×10^{-2}	—
32	1.7309×10^{-2}	1.5336
64	7.9544×10^{-3}	1.1217
128	3.0467×10^{-3}	1.3845
256	1.4243×10^{-3}	1.0971
512	6.5964×10^{-4}	1.1104
1024	2.9541×10^{-4}	1.1590

(b) Second order Godunov scheme

Table 7: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the nonlocal model with the Riemann initial data (19).

Discontinuous solution

Cells	L^1 -errors	OOO
16	1.0844×10^{-1}	–
32	4.4663×10^{-2}	1.2798
64	3.3717×10^{-2}	0.4056
128	1.6571×10^{-2}	1.0248
256	8.2085×10^{-3}	1.0134
512	3.9258×10^{-3}	1.0641
1024	1.7225×10^{-3}	1.1885

(a) First order Godunov scheme

Cells	L^1 -errors	OOO
16	7.1040×10^{-2}	–
32	1.8091×10^{-2}	1.9734
64	1.5403×10^{-2}	0.2321
128	5.7654×10^{-3}	1.4177
256	1.9766×10^{-3}	1.5444
512	5.6019×10^{-4}	1.8190
1024	1.4551×10^{-4}	1.9448

(b) Second order Godunov scheme

Table 8: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the nonlocal model with with the Riemann initial data $u_L = 0.1 < 0.6 = u_R$.

Smooth solution

Cells	L^1 -errors	OOO
16	1.8322×10^{-1}	–
32	8.6063×10^{-2}	1.0901
64	4.5197×10^{-2}	0.9291
128	2.2487×10^{-2}	1.0071
256	1.1061×10^{-2}	1.0237
512	5.1421×10^{-3}	1.1050
1024	2.2161×10^{-3}	1.2143

(a) First order Godunov scheme

Cells	L^1 -errors	OOO
16	7.5224×10^{-2}	–
32	2.2102×10^{-2}	1.7670
64	6.3384×10^{-3}	1.8020
128	2.1914×10^{-3}	1.5322
256	4.8969×10^{-4}	2.1619
512	1.5326×10^{-4}	1.6759
1024	3.0656×10^{-5}	2.3218

(b) Second order Godunov scheme

Table 9: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the nonlocal model with the smooth initial data (21).

5.5 Numerical Convergence to the local Model

Rarefaction solution

Cells	L^1 -errors	OOO
16	2.0744×10^{-1}	–
32	1.4154×10^{-1}	0.5515
64	9.0816×10^{-2}	0.6402
128	5.7350×10^{-2}	0.6632
256	3.5334×10^{-2}	0.6987
512	2.1299×10^{-2}	0.7303
1024	1.2565×10^{-2}	0.7614

(a) First order Godunov scheme

Cells	L^1 -errors	OOO
16	1.6004×10^{-1}	–
32	1.0047×10^{-1}	0.6717
64	6.1193×10^{-2}	0.7153
128	3.6977×10^{-2}	0.7267
256	2.1940×10^{-2}	0.7530
512	1.2857×10^{-2}	0.7711
1024	7.4261×10^{-3}	0.7918

(b) Second order Godunov scheme

Table 10: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the local model with the Riemann initial data (19).

Discontinuous solution

Cells	L^1 -errors	OOC
16	1.2876×10^{-1}	–
32	8.5352×10^{-2}	0.5932
64	4.9126×10^{-2}	0.7969
128	2.6836×10^{-2}	0.8723
256	1.3336×10^{-2}	1.0089
512	6.7539×10^{-3}	0.9815
1024	3.3509×10^{-3}	1.0112

(a) First order Godunov scheme

Cells	L^1 -errors	OOC
16	9.8507×10^{-2}	–
32	6.1071×10^{-2}	0.6897
64	2.9665×10^{-2}	1.0417
128	1.5514×10^{-2}	0.9352
256	7.7374×10^{-3}	1.0036
512	4.0784×10^{-3}	0.9238
1024	1.8805×10^{-3}	1.1169

(b) Second order Godunov scheme

Table 11: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the local model with with the Riemann initial data $u_L = 0.1 < 0.6 = u_R$.

Smooth solution

Cells	L^1 -errors	OOC
16	3.1430×10^{-1}	–
32	1.8983×10^{-1}	0.7275
64	1.0959×10^{-1}	0.7926
128	6.1083×10^{-2}	0.8433
256	3.3015×10^{-2}	0.8876
512	1.7170×10^{-2}	0.9432
1024	8.4186×10^{-3}	1.0283

(a) First order Godunov scheme

Cells	L^1 -errors	OOC
16	2.3215×10^{-1}	–
32	1.3042×10^{-1}	0.8319
64	7.2110×10^{-2}	0.8549
128	3.9305×10^{-2}	0.8755
256	2.1099×10^{-2}	0.8975
512	1.1225×10^{-2}	0.9104
1024	5.8947×10^{-3}	0.9292

(b) Second order Godunov scheme

Table 12: L^1 -errors and observed order of convergence obtained using the the Godunov first and second order scheme in solving the local model with the smooth initial data (21).

5.6 The first Euler forward stage for $g(u) = u$ and $v(u) = 1 - u$

We consider the nonlocal traffic model

$$\partial_t \rho + \partial_x (\rho(1 - \rho * w_\eta)) = 0.$$

Let $\Delta x > 0$ and $\Delta t > 0$ be the space and time steps, and set

$$\lambda := \frac{\Delta t}{\Delta x}, \quad x_j := j\Delta x, \quad t^n := n\Delta t.$$

At time t^n , let ρ_j^n denote the cell average on the cell

$$I_j := \left(x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}\right).$$

Step 1: MUSCL reconstruction and slopes. Following the 2023 scheme, we reconstruct a piecewise linear function

$$\tilde{\rho}^n(x) = \rho_j^n + \frac{x - x_j}{\Delta x} \sigma_j^n, \quad x \in I_j,$$

where the slope is chosen by the minmod limiter

$$\sigma_j^n = \minmod\left(\rho_j^n - \rho_{j-1}^n, \frac{1}{2}(\rho_{j+1}^n - \rho_{j-1}^n), \rho_{j+1}^n - \rho_j^n\right).$$

Here

$$\minmod(a_1, \dots, a_m) = \begin{cases} \operatorname{sgn}(a_1) \min_{1 \leq \ell \leq m} |a_\ell|, & \text{if } \operatorname{sgn}(a_1) = \dots = \operatorname{sgn}(a_m), \\ 0, & \text{otherwise.} \end{cases}$$

The interface values are

$$\rho_{j+\frac{1}{2},-}^n = \rho_j^n + \frac{1}{2}\sigma_j^n, \quad \rho_{j+\frac{1}{2},+}^n = \rho_{j+1}^n - \frac{1}{2}\sigma_{j+1}^n.$$

Step 2: Arbitrary- η trapezoidal quadrature and normalization.

$$r := \left\lceil \frac{\eta}{\Delta x} \right\rceil$$

We denote the left endpoint quadrature $w_k := \omega_\eta(k\Delta x)$ for $k = 0, \dots, r$, and normalize the trapezoidal quadrature as in Remark 3.2 of the 2023 paper, by setting

$$Q_k := \sum_{k=0}^{r-1} \frac{\Delta x}{2} (w_k + w_{k+1}), \quad \tilde{w}_k := \frac{w_k}{Q_k}.$$

Then

$$\sum_{k=0}^{r-1} \frac{\Delta x}{2} (\tilde{w}_k + \tilde{w}_{k+1}) = 1.$$

Define the weights

$$\alpha_k := \frac{\Delta x}{2} \tilde{w}_k, \quad \beta_k := \frac{\Delta x}{2} \tilde{w}_{k+1}, \quad k = 0, \dots, r-1.$$

Hence

$$\alpha_k \geq 0, \quad \beta_k \geq 0, \quad \sum_{k=0}^{r-1} (\alpha_k + \beta_k) = 1.$$

Step 3: The downstream average. The convulsion is approximated by

$$R_{j+\frac{1}{2}}^n = \sum_{k=0}^{r-1} \alpha_k \rho_{j+k+\frac{1}{2},+}^n + \sum_{k=0}^{r-1} \beta_k \rho_{j+k+\frac{3}{2},-}^n \quad (26)$$

Step 4: The first Euler forward stage. Since $g(u) = u$ and $v(u) = 1 - u$, the Godunov numerical flux becomes

$$f_{j+\frac{1}{2}}^n = \rho_{j+\frac{1}{2},-}^n \left(1 - R_{j+\frac{1}{2}}^n \right).$$

Therefore the first Euler forward stage is

$$\rho_j^{(1)} = \rho_j^n - \lambda \left(f_{j+\frac{1}{2}}^n - f_{j-\frac{1}{2}}^n \right), \quad (27)$$

that is,

$$\rho_j^{(1)} = \rho_j^n - \lambda \left(\rho_{j+\frac{1}{2},-}^n (1 - R_{j+\frac{1}{2}}^n) - \rho_{j-\frac{1}{2},-}^n (1 - R_{j-\frac{1}{2}}^n) \right). \quad (28)$$

5.7 Transformation to the 2021-style form

Define the two-point flux

$$\mathcal{G}(a, b) := a(1 - b).$$

Also set

$$a_j^n := \rho_{j+\frac{1}{2},-}^n.$$

Using (26), we compute

$$\begin{aligned} f_{j+\frac{1}{2}}^n &= a_j^n \left(1 - R_{j+\frac{1}{2}}^n \right) \\ &= a_j^n \left(1 - \sum_{k=0}^{r-1} \alpha_k \rho_{j+k+\frac{1}{2},+}^n - \sum_{k=0}^{r-1} \beta_k \rho_{j+k+\frac{3}{2},-}^n \right). \end{aligned} \quad (29)$$

Since

$$\sum_{k=0}^{r-1} (\alpha_k + \beta_k) = 1,$$

we may rewrite the factor 1 as

$$1 = \sum_{k=0}^{r-1} (\alpha_k + \beta_k).$$

Substituting this into (29), we obtain

$$\begin{aligned} f_{j+\frac{1}{2}}^n &= \sum_{k=0}^{r-1} \alpha_k a_j^n \left(1 - \rho_{j+k+\frac{1}{2},+}^n\right) + \sum_{k=0}^{r-1} \beta_k a_j^n \left(1 - \rho_{j+k+\frac{2}{3},-}^n\right) \\ &= \sum_{k=0}^{r-1} \alpha_k \mathcal{G}\left(a_j^n, \rho_{j+k+\frac{1}{2},+}^n\right) + \sum_{k=0}^{r-1} \beta_k \mathcal{G}\left(a_j^n, \rho_{j+k+\frac{2}{3},-}^n\right). \end{aligned} \quad (30)$$

This shows that the flux $f_{j+\frac{1}{2}}^n$ is a convex combination of two-point flux values, because all coefficients are nonnegative and their sum is equal to one.

Substituting (30) into (27), the first Euler forward stage becomes

$$\begin{aligned} \rho_j^{(1)} &= \rho_j^n - \lambda \sum_{k=0}^{r-1} \alpha_k \left[\mathcal{G}\left(a_j^n, \rho_{j+k+\frac{1}{2},+}^n\right) - \mathcal{G}\left(a_{j-1}^n, \rho_{j+k-\frac{1}{2},+}^n\right) \right] \\ &\quad - \lambda \sum_{k=0}^{r-1} \beta_k \left[\mathcal{G}\left(a_j^n, \rho_{j+k+\frac{2}{3},-}^n\right) - \mathcal{G}\left(a_{j-1}^n, \rho_{j+k+\frac{1}{2},-}^n\right) \right]. \end{aligned} \quad (31)$$

This is the desired representation: the update is a weighted sum of two-point flux differences, with nonnegative weights inherited from the normalized trapezoidal quadrature.

To make the convex-combination structure completely explicit, define

$$\gamma_{2k} := \alpha_k, \quad \gamma_{2k+1} := \beta_k, \quad k = 0, \dots, r-1,$$

and

$$X_{2k}^n := \rho_{j+k+\frac{1}{2},+}^n, \quad X_{2k+1}^n := \rho_{j+k+\frac{2}{3},-}^n.$$

Then

$$\gamma_m \geq 0, \quad \sum_{m=0}^{2r-1} \gamma_m = 1,$$

and (30) becomes

$$f_{j+\frac{1}{2}}^n = \sum_{m=0}^{2r-1} \gamma_m \mathcal{G}(a_j^n, X_m^n). \quad (32)$$

Hence the Euler forward stage can be written as

$$\rho_j^{(1)} = \rho_j^n - \lambda \sum_{m=0}^{2r-1} \gamma_m \left[\mathcal{G}(a_j^n, X_m^n) - \mathcal{G}(a_{j-1}^n, X_{m-1}^n) \right]. \quad (33)$$

Formula (32) shows directly that $f_{j+\frac{1}{2}}^n$ is a convex combination of the two-point flux values

$$\mathcal{G}(a_j^n, X_m^n), \quad m = 0, \dots, 2r-1,$$

and (33) is the flattened 2021-style form of the first Euler forward stage.

$$r := \lfloor \frac{\eta a}{\Delta x} \rfloor$$

6 Convergence of the numerical method to entropy solution

References

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